Project Title

News & Social Media Analytics Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
Harshitha Bollineni(2520080007)
Naeeni Satwika(2520090151)

Team Number: 25

Section: 12

Under the guidance of

Ms.Chandusha Kanda

Assistant professor, CSIT



Hyderabad-500090, Telangana, India.

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY/ARTIFICIAL INTELLIGENCE & DATA SCIENCE



DECLARATION

The Mini Project Report entitled "**News & Social Media Analytics Engine"** is a record of Bonafide work of Harshitha Bollineni – 2520080007., Naeeni Satwika- 2520090151 submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering/Computer Science and Information Technology/ Artifical Intelligence & data Science to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

Harshitha Bollineni– 2520080007

Naeeni Satwika– 2520090151

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



CERTIFICATE

This is to certify that the mini project based report entitled "**News & Social Media Analytics Engine"** is a bonafide work done and submitted by Harshitha Bollineni – 2520080007., Naeeni Satwika- 2520090151 in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering/ Computer Science and Information Technology/Artificial Intelligence & Data Science, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, Ms.Chandusha Kanda, Assistant Professor, CSIT which cannot be expressed in words for his/her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. P.Venkateshwar Rao (CSE)/Dr. Khaja Shareef (CSIT)/ Dr. Sreenath Itikela (AI&DS)**, Head of the *Department for Computer Science Engineering/ Department for Computer Science and Information Technology/Department of Artificial Intelligence and Data Science* for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

Harshitha Bollineni– 2520080007

Naeeni Satwika– 2520090151

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1.ABSTRACT

The **News & Social Media Analytics Engine** is a Data Structures and Algorithms (DSA)-based platform designed to efficiently process, search, organize, and analyze large volumes of news articles and social media posts. The project addresses major challenges such as information overload, duplicate content, related-story identification, inefficient search, and the need for scalable processing. The proposed system integrates multiple advanced algorithms into a unified analytics pipeline. **KMP and Rabin-Karp algorithms** are used for fast pattern searching, while **Suffix Arrays and Longest Common Prefix (LCP)** techniques support document similarity and repeated-content detection. **Dynamic Programming** is applied for fuzzy matching and query correction, improving the accuracy and usability of search operations. In addition, **network-flow techniques** support moderator assignment and resource allocation, while randomized algorithms and parallel processing improve scalability for large workloads. The system follows an end-to-end pipeline involving content collection, normalization, searching, comparison, correction, clustering, assignment, and analytics. The proposed platform aims to reduce information redundancy, improve content organization and retrieval, enable efficient moderation, and provide scalable analytics through an interactive dashboard. Overall, the project demonstrates the practical integration of advanced DSA techniques to build an efficient and scalable platform for large-scale digital content analysis.

Team Name: News & Social Media Analytics Engine

Team Logo (if any):

Team Members:

1. **Harshitha Bollineni– 2520080007**
2. **Naeeni Satwika– 2520090151**

1. INTRODUCTION

The rapid growth of digital platforms has resulted in an enormous amount of **news articles and social media content** being generated every day. While this provides users with access to information from multiple sources, it also creates significant challenges such as information overload, duplicate content, difficulty in finding relevant information, identification of related stories, and efficient management of large-scale textual data. Manually searching, comparing, and organizing such content becomes increasingly difficult as the volume of information grows. dsa ppt - news and social media

The **News & Social Media Analytics Engine** is proposed as a DSA-based solution to address these challenges by combining multiple efficient data structures and algorithms within a single analytics platform. The project focuses on fast pattern searching using **KMP and Rabin-Karp**, document similarity and repeated-content analysis using **Suffix Arrays and LCP**, and fuzzy matching and query correction using **Dynamic Programming**. Network-flow techniques are incorporated to support moderator assignment and resource allocation, while randomized algorithms and parallel processing are considered for improving scalability and performance when handling large workloads. dsa ppt - news and social media

The system follows an end-to-end processing pipeline that begins with collecting news articles and social media posts and proceeds through **normalization, pattern searching, similarity comparison, query correction, story clustering, moderator assignment, and analytics**. The processed information is presented as organized and searchable analytics through the user interface. dsa ppt - news and social media

This project demonstrates how advanced DSA concepts can be practically integrated to build an efficient and scalable platform for digital content analysis. The following sections present the **technical background, methodology, algorithms, system architecture, expected outcomes, and future scope** of the proposed News & Social Media Analytics Engine. dsa ppt - news and social media

3. LITERATURE SURVEY

1. GDELT Project:

GDELT is a large-scale global news monitoring system that processes news from many sources and organizes information about events, entities, and topics. It demonstrates large-scale news collection and analysis. This is similar to our project's objective of processing large volumes of digital content. Our project additionally focuses on DSA-based searching, similarity detection, clustering, and assignment. GDELT Cloud

2. GDELT Cloud:

GDELT Cloud processes news articles and groups related articles into coherent stories while identifying duplicate coverage. It provides a useful reference for story clustering and redundancy reduction. These functions are closely related to our duplicate detection and story-clustering modules. Our system approaches these operations using dedicated DSA techniques such as suffix arrays and LCP. GDELT Cloud

3. News API:

News API provides an interface for searching and retrieving news articles from multiple sources. It demonstrates efficient collection and retrieval of current news information. This is relevant to the content collection and search stages of our system. Our project extends this concept by applying pattern-matching algorithms and similarity analysis to the collected content.

4. News Near-Duplicate Detection:

This project detects highly similar news articles using character shingles, MinHash, and Locality Sensitive Hashing (LSH). It focuses on identifying redundant content in large news collections. The objective is similar to our duplicate-detection module. Our project uses suffix arrays and LCP as an alternative DSA-based approach for similarity and repeated-content analysis.

5. News Clustering and Summarization:

This project groups similar news articles using sentence embeddings and clustering algorithms such as K-Means and DBSCAN. It helps users organize large volumes of news and generate concise summaries. It is related to our story-clustering objective. Our system differs by integrating classical DSA algorithms with searching, similarity analysis, and query correction.

6. Real-Time News Clustering:

The News Clustering project tracks developing stories across different news outlets and groups articles according to semantic similarity. It provides a timeline for observing how stories develop

over time. This supports the need for efficient organization of related news. Our project similarly clusters stories while incorporating KMP, Rabin-Karp, suffix arrays, LCP, and other DSA techniques.

7. Similar News Detection:

This project identifies similar news articles by preprocessing text, generating embeddings, clustering articles, and calculating cosine similarity. It reduces the number of comparisons required during similarity searching. This is relevant to our document-comparison and retrieval objectives. Our project uses DSA-based similarity and pattern-processing techniques rather than relying only on embeddings.

8. News Pulse:

News Pulse is a full-stack news aggregation system that collects articles from RSS feeds, removes duplicates, and groups articles covering the same story into clusters. It combines data collection, storage, clustering, and visualization. These functions are similar to our proposed analytics pipeline. Our project further incorporates query correction, moderator assignment, and scalable DSA processing.

References

1. GDELT Project — https://www.gdelt.org/
2. GDELT Cloud — https://gdeltcloud.com/
3. News API — https://newsapi.org/

4. Problem Statement and Objectives

Problem Statement:

The rapid growth of news and social media content makes it difficult to efficiently search, identify duplicate information, and organize related digital content.

Problem Description:

Large volumes of news articles and social media posts require efficient techniques for searching, duplicate detection, story clustering, and query correction. The project uses Data Structures and Algorithms to make content analysis faster and more organized.

Objectives:

- To enable fast searching of patterns in large text collections using KMP and Rabin-Karp algorithms.
- To detect duplicate and similar content using Suffix Arrays and LCP.
- To group related news articles and social media posts into meaningful story clusters.
- To provide query correction and fuzzy matching using Dynamic Programming.
- To support efficient moderator assignment and resource allocation using Network Flow.
- To improve the scalability and efficiency of processing large volumes of digital content.

5.HARDWARE AND SOFTWARE REQUIREMENTS

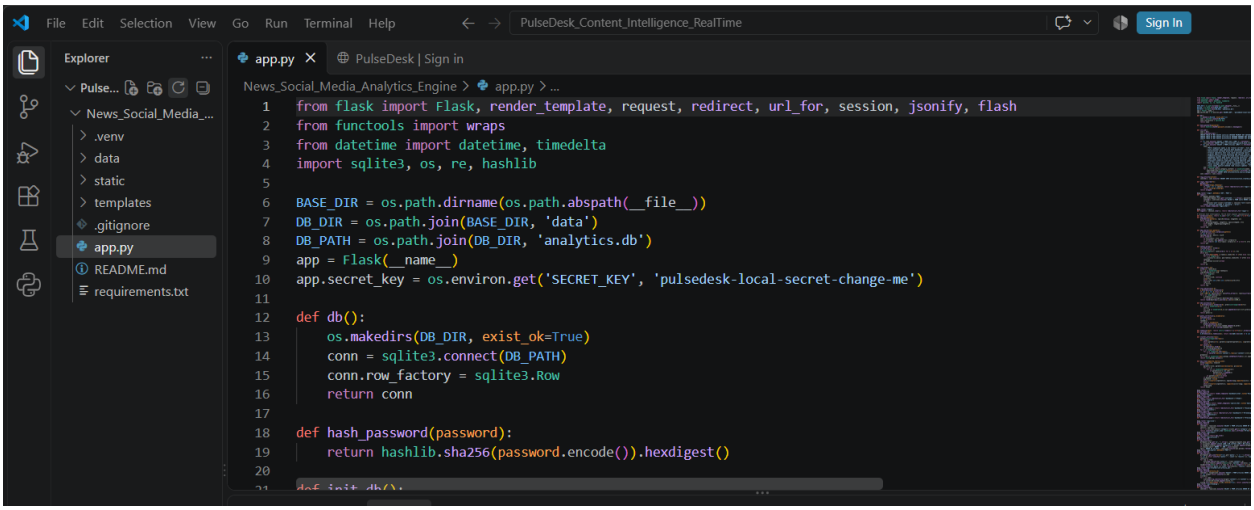
Hardware Requirements:

- Intel Core i5 or equivalent processor
- 8 GB RAM
- 10 GB free storage
- 1366 × 768 or higher display
- Internet connection

Software Requirements:

- Windows 10/11 or equivalent OS
- Python 3.x
- Flask 3.x
- SQLite Database
- HTML, CSS, JavaScript
- Visual Studio Code
- Google Chrome / Microsoft Edge

6.IMPLEMENTATION

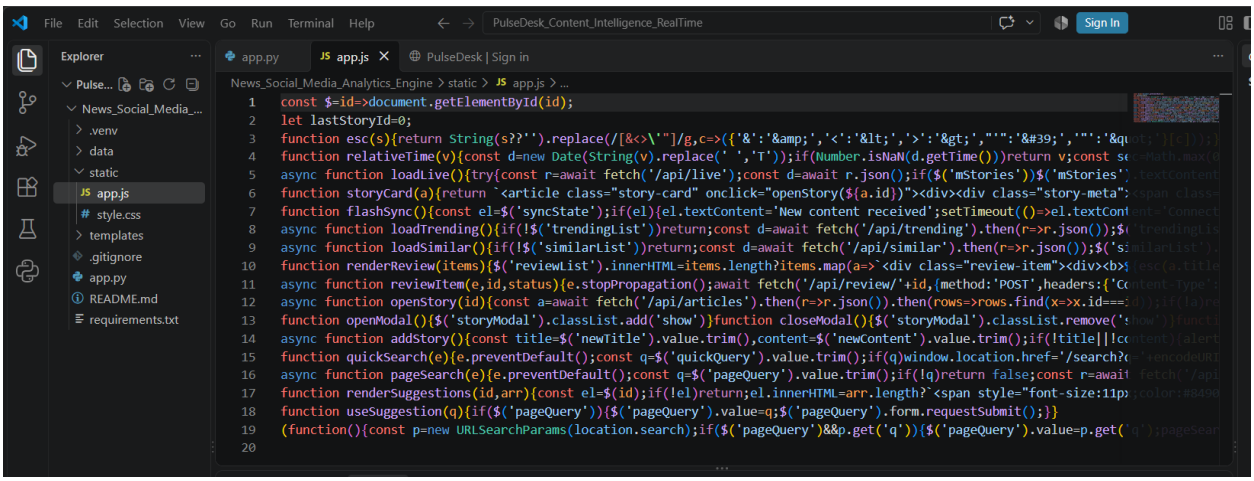


```python
from flask import Flask, render_template, request, redirect, url_for, session, jsonify, flash
from functools import wraps
from datetime import datetime, timedelta
import sqlite3, os, re, hashlib

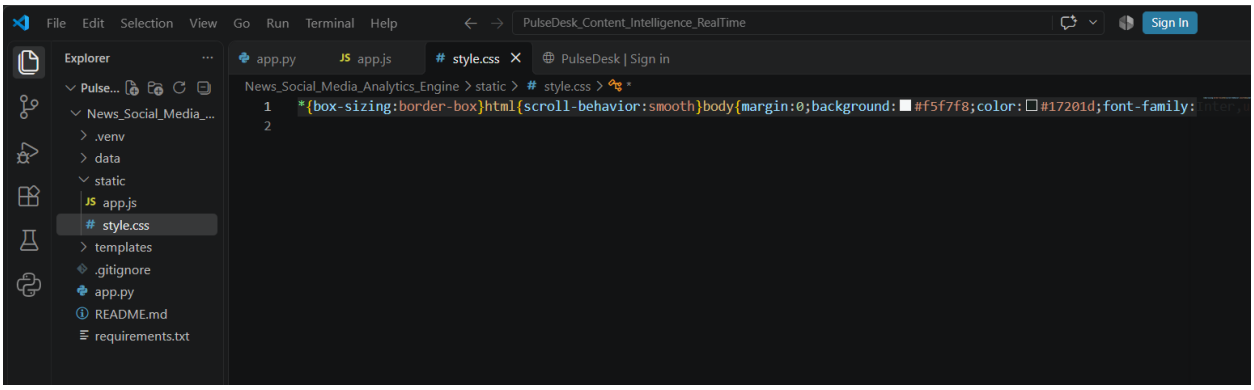
BASE_DIR = os.path.dirname(os.path.abspath(__file__))
DB_DIR = os.path.join(BASE_DIR, 'data')
DB_PATH = os.path.join(DB_DIR, 'analytics.db')
app = Flask(__name__)
app.secret_key = os.environ.get('SECRET_KEY', 'pulsedesk-local-secret-change-me')

def db():
    os.makedirs(DB_DIR, exist_ok=True)
    conn = sqlite3.connect(DB_PATH)
    conn.row_factory = sqlite3.Row
    return conn

def hash_password(password):
    return hashlib.sha256(password.encode()).hexdigest()
```



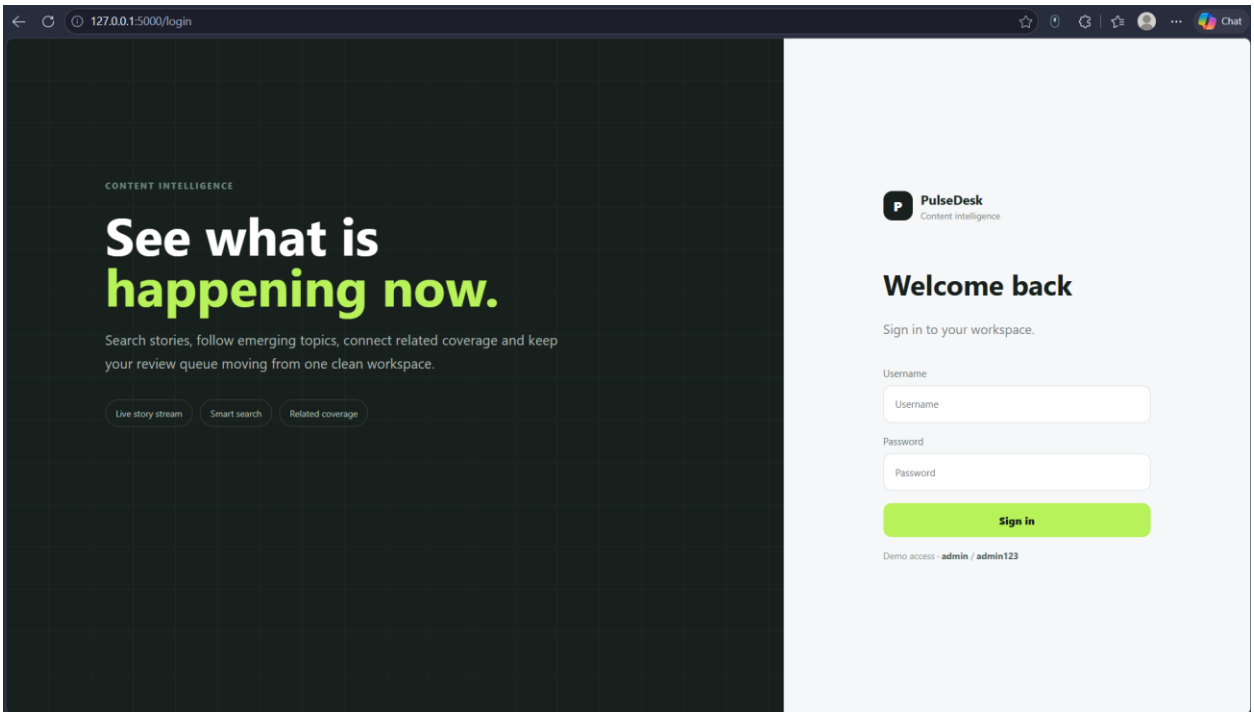
```javascript
const $=id=>document.getElementById(id);
let lastStoryId=0;
function esc(s){return String(s??'').replace(/[&<>\'"]/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&#39;',"'":'&#39;'}[c]));}
function relativeTime(v){const d=new Date(String(v).replace(' ','T'));if(Number.isNaN(d.getTime()))return v;const s
async function loadLive(){try{const r=await fetch('/api/live');const d=await r.json();if($('mStories'))$('mStories'
function storyCard(a){return `<article class="story-card" onclick="openStory(${a.id})"><div><div class="story-meta"
function flashSync(){const el=$('syncState');if(el){el.textContent='New content received';setTimeout(()=>el.textCon
async function loadTrending(){if(!$('trendingList'))return;const d=await fetch('/api/trending').then(r=>r.json());$(
async function loadSimilar(){if(!$('similarList'))return;const d=await fetch('/api/similar').then(r=>r.json());$('s
function renderReview(items){$('reviewList').innerHTML=items.length?items.map(a=>`<div class="review-item"><div><b>
async function reviewItem(e,id,status){e.stopPropagation();await fetch('/api/review/'+id,{method:'POST',headers:{'C
async function openStory(id){const a=await fetch('/api/articles').then(r=>r.json()).then(rows=>rows.find(x=>x.id===
function openModal(){$('storyModal').classList.add('show')}function closeModal(){$('storyModal').classList.remove('
async function addStory(){const title=$('newTitle').value.trim(),content=$('newContent').value.trim();if(!title||!c
function quickSearch(e){e.preventDefault();const q=$('quickQuery').value.trim();if(q)window.location.href='/search?
async function pageSearch(e){e.preventDefault();const q=$('pageQuery').value.trim();if(!q)return false;const r=awai
function renderSuggestions(id,arr){const el=$(id);if(!el)return;el.innerHTML=arr.length?`<span style="font-size:11p
function useSuggestion(q){if($('pageQuery')){$('pageQuery').value=q;$('pageQuery').form.requestSubmit();}}
(function(){const p=new URLSearchParams(location.search);if($('pageQuery')&&p.get('q')){$('pageQuery').value=p.get('
```


File Edit Selection View Go Run Terminal Help
PulseDesk_Content_Intelligence_RealTime
Sign In
Explorer
app.py
app.js
style.css
PulseDesk | Sign in
News_Social_Media_Analytics_Engine > static > style.css
1 *{box-sizing:border-box}html{scroll-behavior:smooth}body{margin:0;background:#f5f7f8;color:#17201d;font-family:
2
Pulse...
News_Social_Media_...
.venv
data
static
app.js
style.css
templates
.gitignore
app.py
README.md
requirements.txt

2. RESULTS & DISCUSSION

The developed News & Social Media Analytics Engine successfully provides a normal-user web interface for managing and analyzing news and social-media-style content. The application integrates searching, related-story detection, duplicate detection, trending topics, review management, and new-story publishing into a single platform.

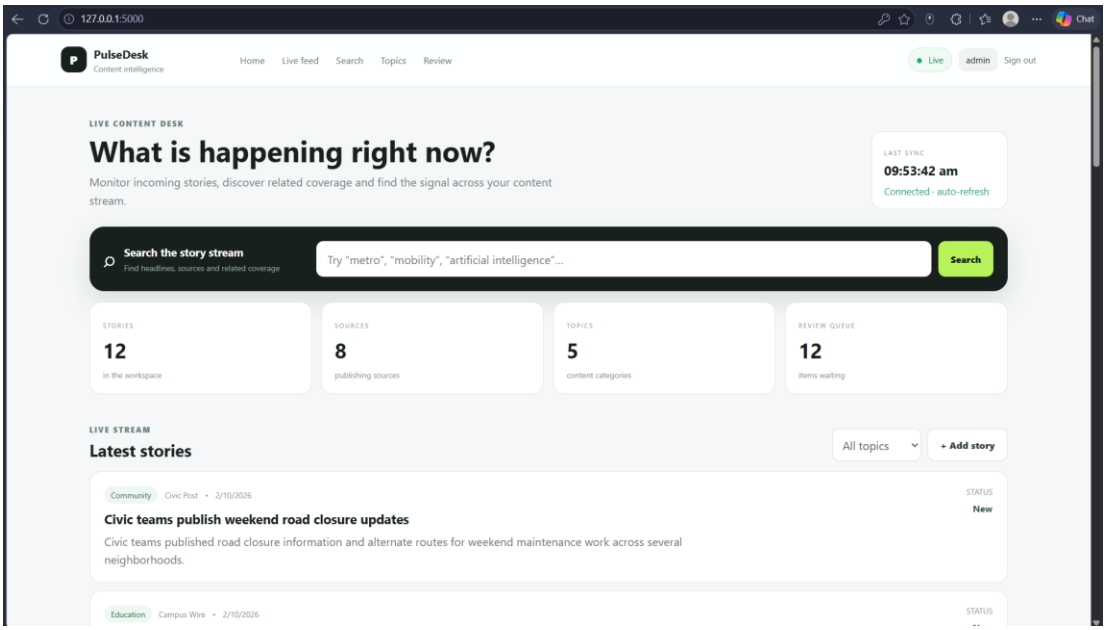
7.1 Login Screen



The login screen provides secure access to the News & Social Media Analytics Engine using registered user credentials.

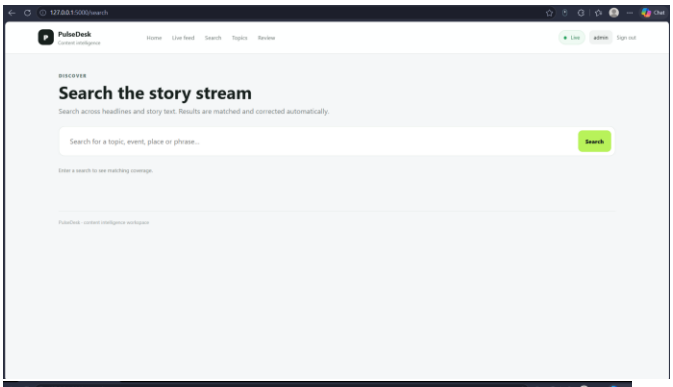
After successful authentication, the user is redirected to the main dashboard.

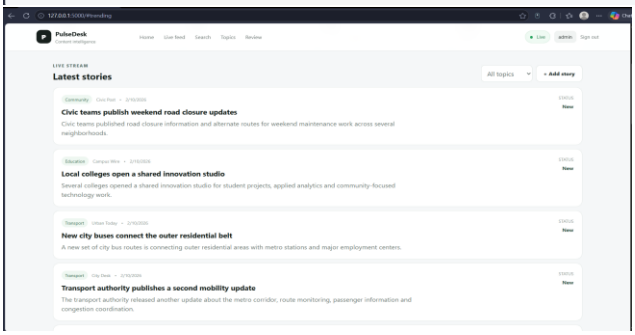
7.2 Main Dashboard / Live Story Feed



The application automatically refreshes the content stream to provide an updated user experience. The dashboard displays the available news stories in a user-friendly live-style content stream.

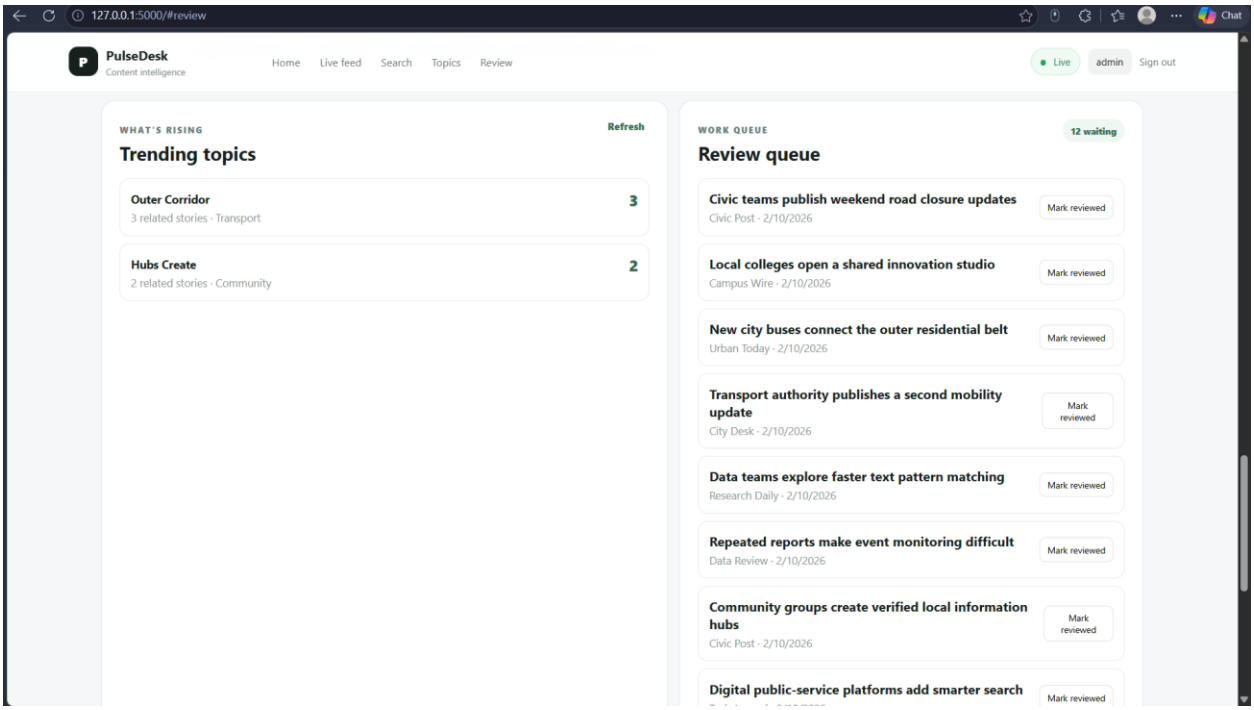
7.3 Smart Search & Related stories





The system retrieves relevant articles based on the user's search query and provides suggestions for incorrect queries.It also identifies related or repeated stories using content similarity analysis.

7.3 Trending Topics and Review Queue



The system groups similar articles to identify trending topics and repeated coverage.

The review queue helps organize articles that require checking and supports moderator workload assignment.

8. CONCLUSION & FUTURE SCOPE

The **News & Social Media Analytics Engine** successfully demonstrates the practical application of advanced Data Structures and Algorithms to news and social media content analysis. The system provides efficient searching, identification of duplicate and related content, article clustering, query correction, and moderator workload assignment. By integrating these techniques into a web-based platform, the project helps reduce information redundancy and improves the organization and retrieval of digital content.

The implementation demonstrates that advanced DSA concepts can be effectively applied to a practical content-management problem. Overall, the project meets its objectives of improving content analysis, retrieval, organization, and management while providing an efficient and scalable foundation for handling large volumes of textual data.

FUTURE SCOPE

The system can be further enhanced in the following stages:

Near-Term Scope

- Improve the accuracy and efficiency of **search, similarity detection, and query correction** for larger collections of news and social-media content.
- Enhance the **article clustering and duplicate detection** features to provide more accurate grouping of related stories.
- Improve the user interface and content-processing workflow for smoother handling of large datasets.

Medium-Term Scope

- Integrate **Rabin-Karp** along with the existing pattern-searching techniques for more flexible and efficient text searching.
- Extend the moderator assignment mechanism using more advanced **network-flow and resource-allocation techniques**.
- Support larger and continuously growing datasets for improved content analysis and retrieval.

Long-Term Scope

- Incorporate **parallel processing** to improve performance when processing very large volumes of content.
- Explore **randomized algorithms** for improved scalability and efficient large-scale processing.

- Extend the system toward a more intelligent and scalable platform capable of providing **real-time analytics** for large-scale news and social-media content.

9. REFERENCES

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   SIAM Digital Library – Fast Pattern Matching in Strings
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   SIAM Digital Library – Suffix Arrays
3. **SQLite Documentation**, SQLite Consortium.
   Used as a reference for the database component of the project, which stores users, articles, and activity data.
   SQLite Documentation
4. **SQLite SQL Language Documentation**, SQLite Consortium.
   Used for database operations and SQL-based storage and retrieval.
   SQLite SQL Language Documentation
5. **Flask Documentation**, Pallets Projects.
   Used as the reference for developing the Python-based web application and Flask routing used in the project.
   Flask Documentation
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7. This provides the theoretical basis for edit-distance-based fuzzy matching used for query correction.
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15. Internal project reference containing the proposed system objectives, algorithms, and implementation details.